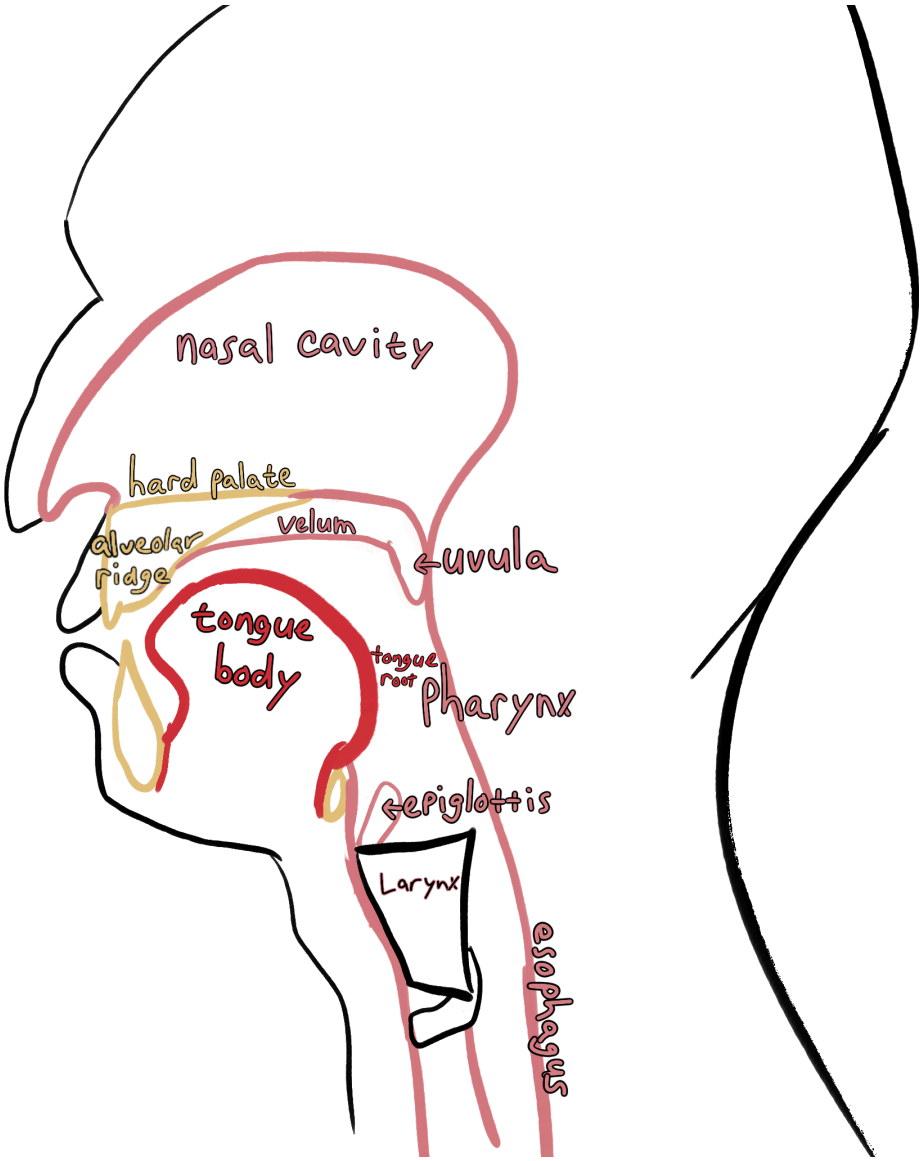


Speech Anatomy 4

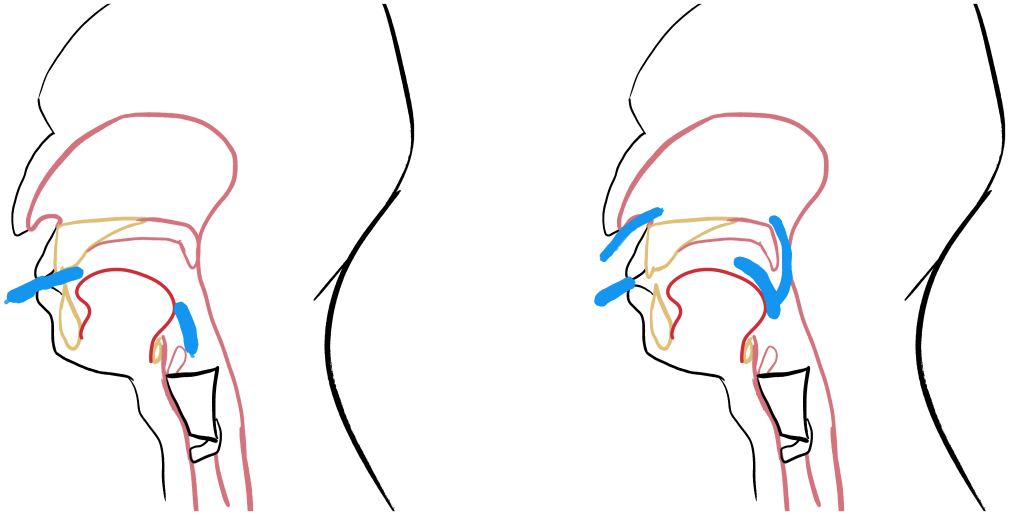
Supralaryngeal Vocal Tract

Dr. Josef Fruehwald

2026-09-01


nasal cavity
hard palate
alveolar ridge
velum
uvula
tongue body
tongue root
pharynx
epiglottis
Larynx
esophagus

Nasopharynx



The Tongue



The Tongue

- **extrinsic** muscles: muscles attached to the tongue
- **intrinsic** muscles: muscles within the tongue

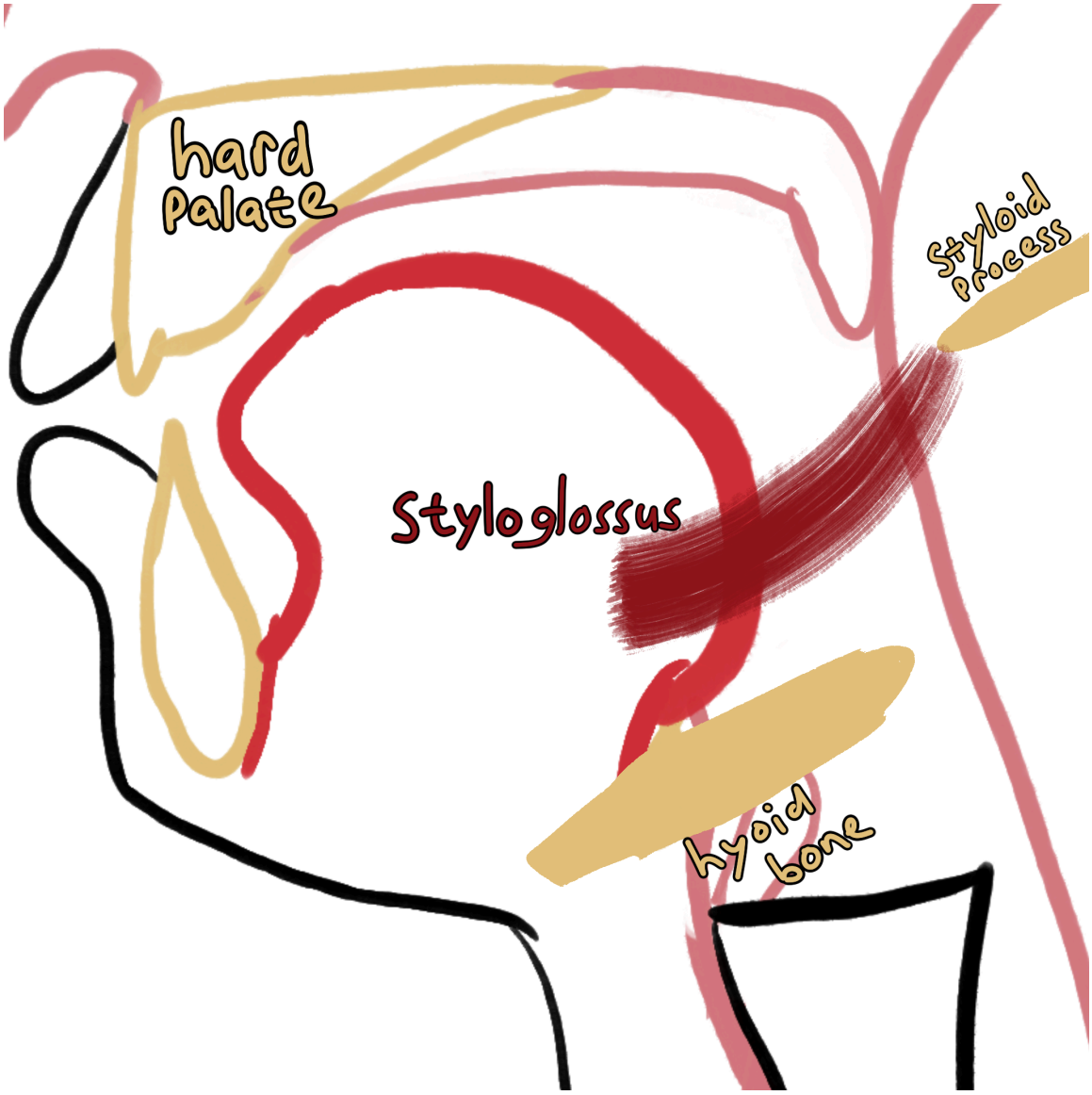
The Tongue

Muscles are usually named for the what they attach to.

- “-glossus”: Attaching to the tongue

The Tongue

Extrinsic Muscles



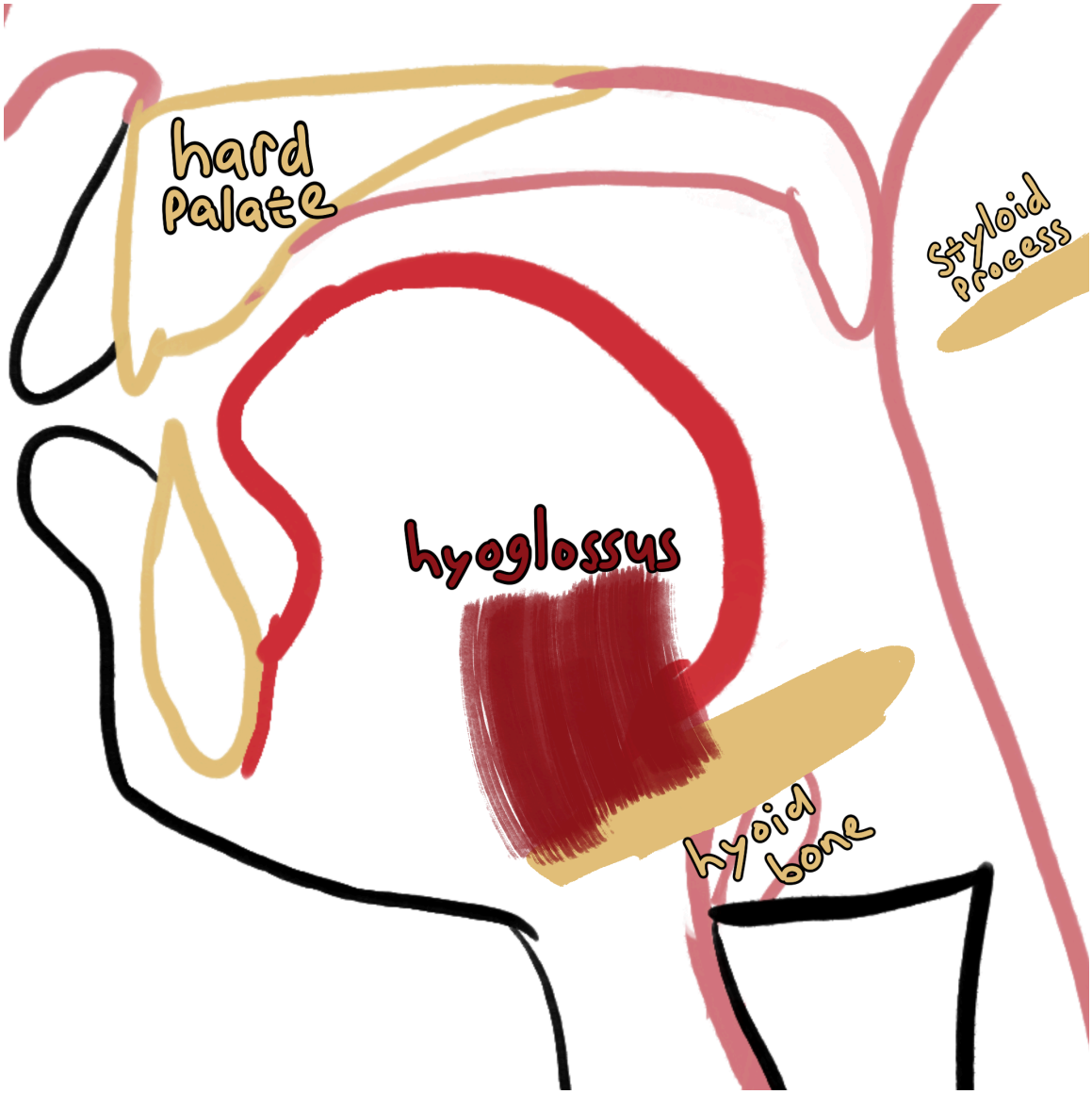
Pulls tongue body back

Styloid Process



The Tongue

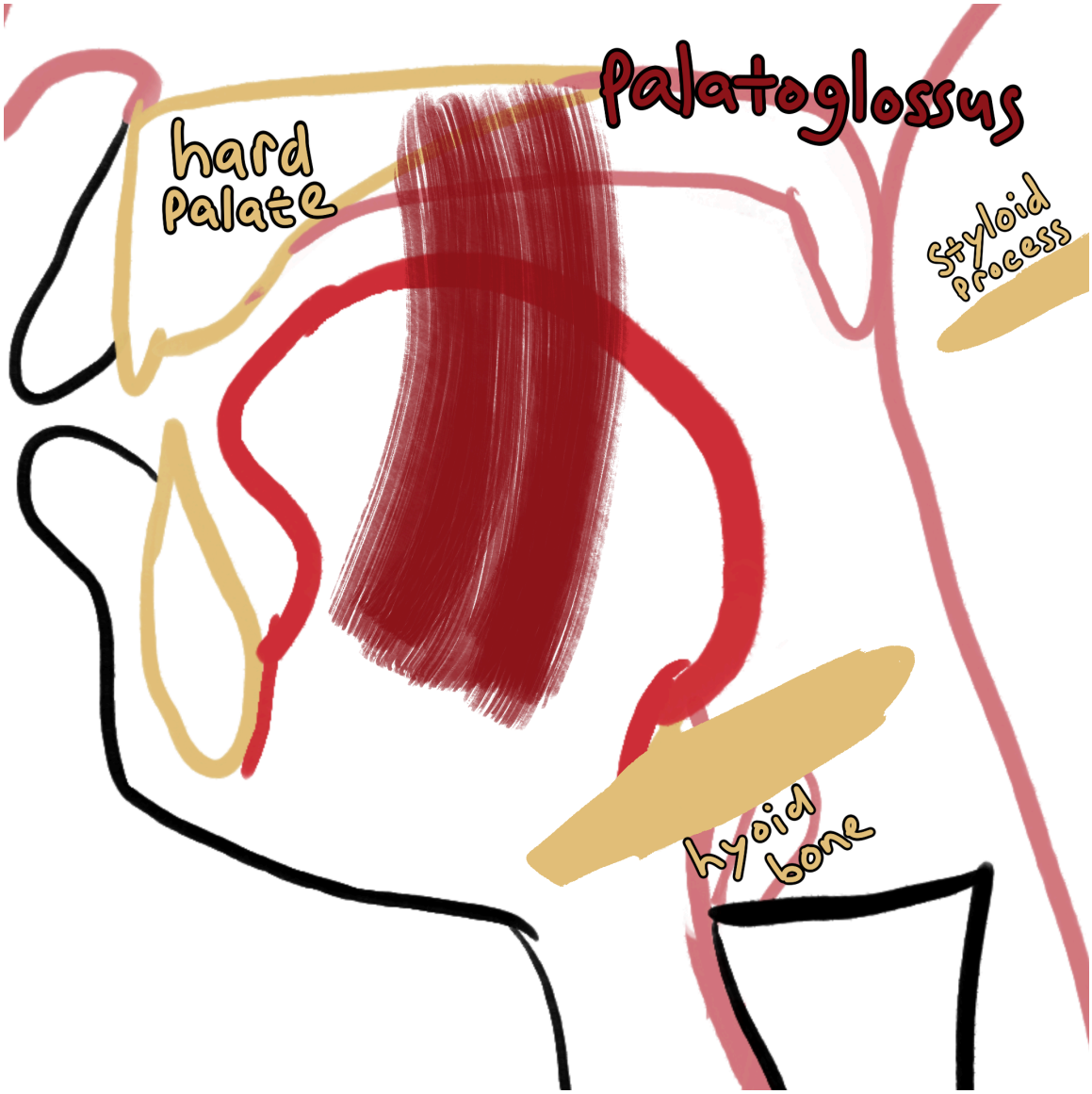
Extrinsic Muscles



Pulls tongue body down

The Tongue

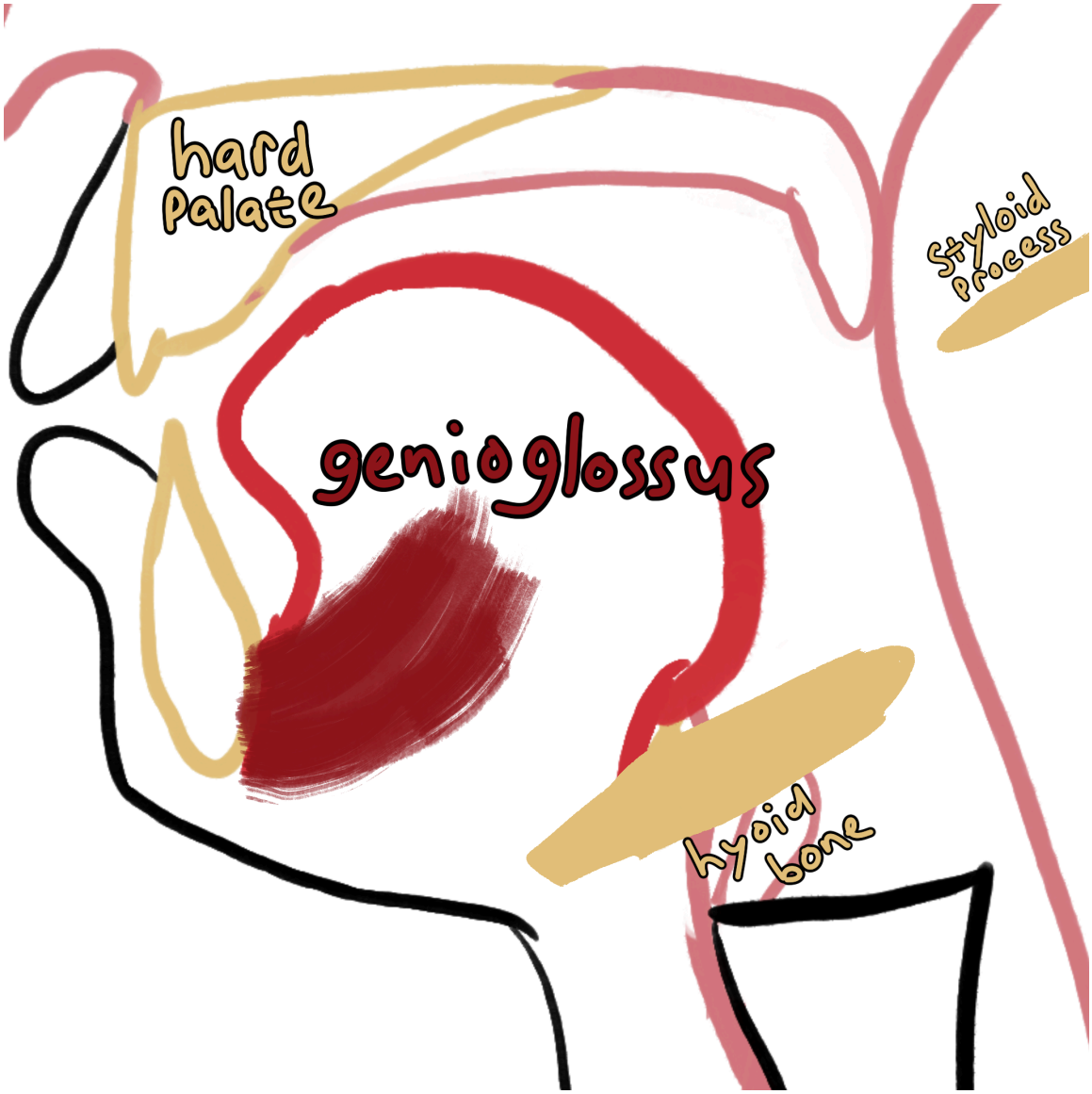
Extrinsic Muscles



Pulls tongue up

The Tongue

Extrinsic Muscles



Downward and forward

Coordinating it all

Everything is connected to everything

- Hyoglossus pulls down on tongue ?
- Tongue pulls on palatoglossus ?

- Palatoglossus pulls on soft palate

Low vowels are more likely to be nasalized than high vowels.

Coordinating it all

Timing

